Commentary2Highlight

Project status and next steps | 19 October 2026

1. Pipeline implementation status

Stage	Status	What works / what remains
Audio and timestamped ASR	Working, with input limits	FFmpeg extracts audio; Faster-Whisper small transcribes and detects language with estimated timestamps. Whisper translate can produce English. Check every video has decodable commentary audio.
Text preparation and NLP cues	Working baseline	Project normalization and event phrases retain negation, hypothetical, retrospective, live-action, and emphasis cues. Translation/ASR errors still affect text.
Candidate events and frames	Partial	Rules propose goal, penalty/VAR, card, shot/chance, and set pieces. Lexical frames are not full semantic-role labeling and do not robustly extract actors/outcomes.
Project-trained DL ranker	Implemented; experimental	Project-fitted TF-IDF feeds a PyTorch MLP trained from random initialization. Multi-task MLP and BiGRU with temporal attention exist but are experimental; the TF-IDF MLP is the current primary ranker.
Entities and coreference	Partial	Exact aliases and cautious short-window pronoun links work if a verified roster is supplied. No trained player-name or nickname model.
Peak grouping and exports	Working baseline	Nearby candidates are grouped and exported as timestamped Markdown, CSV, and JSON. No edited video clips yet; reported timestamps need review.
Evaluation	Needs more independent labels	Whole-match-held-out exploratory evaluation spans seven matches, but mixes weak SoccerNet alignment and limited human review. It does not prove broad highlight quality.

2. NLP and deep-learning components

Pretrained input models

Faster-Whisper small handles multilingual speech recognition, language detection, and timestamps; English translation can use Whisper translate. NLLB-200 distilled 600M is an optional text-translation route. These are pretrained input tools, not team-trained models.

Team-built NLP

Normalization, football event phrase rules, transparent discourse cues, TF-IDF feature fitting, and event-frame templates are project-written methods.

Team-trained DL

The PyTorch MLP learns event-related vs routine scores from project-fitted TF-IDF features. A multi-task MLP adds broad event-family predictions. A BiGRU with temporal attention processes ordered transcript windows. These models are trained from project data; Whisper and NLLB are not fine-tuned.

Exploratory evaluation

Project notes report seven-match whole-match-held-out mean F1 of 0.655 for the TF-IDF MLP and 0.553 for the TF-IDF linear baseline; the experimental sequence model reached 0.597 eventness macro-F1. The labels are mixed and limited, so these figures are exploratory.

3. Demonstrated results and limits

• **Crystal Palace vs Manchester City:** the three-minute sequence demo produced 81 transcript segments and two ranked candidates near 00:01:23 and 00:02:50. The commentary and timestamps are prototype outputs and require review.

• **Language coverage:** examples exist for Spanish, French, and Russian. Translation and player-name spelling vary; this has not been validated across all languages or broadcasters.

• **Audio inventory:** five SoccerNet MKVs are recorded; three have usable audio and two Manchester City-Newcastle files decode to no audio samples.

• **Still incomplete:** robust coreference, full semantic-role labeling, learned player identity/nicknames, visual event detection, automatic video cutting, and a broad cross-language benchmark.

4. Next steps

1. Finish human labels for event candidates and routine controls; check duplicates before adding them to training.

2. Reserve a complete unseen match for final evaluation and add more independent matches/languages.

3. Separate event-related commentary labels from subjective highlight labels; improve missed-event and false-positive review.

4. Build verified rosters and alias lists for selected matches, then measure name correction errors.

5. Compare transcription/translation routes against source audio and verify timestamp offsets.

6. Keep the TF-IDF MLP as primary until another model performs better on held-out matches; treat the multi-task and sequence models as experiments.

5. Summary for the team

“We have a prototype that extracts audio, creates timestamped multilingual transcripts, identifies event-related commentary, and ranks candidate moments with our own NLP features and PyTorch models trained from project data. Whisper and optional translation are pretrained input tools. The current output is timestamped highlight commentary; automatic video cutting is future work. Better labels, cross-match evaluation, player-name handling, and timestamp checks remain.”